Supplementary Materials for

Leveraging Deep Reinforcement Learning within Optimal Renewable Energy Strategies for Sustainable AI Data Centers

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S1. Details of the AI Workloads and AI-focused Data Center Model

This section provides detailed information for supporting the **AI Workloads and AI-focused Data Center Model** Section in the manuscript. AI-focused data centers differ fundamentally from traditional data centers, particularly in power density and computing workloads. Conventional data centers, predominantly utilizing CPU-based servers, typically operate with a maximum power density of about 10,000 W/m². In contrast, AI data centers rely on GPU-intensive servers designed for training and running extreme large AI models, exhibiting significantly higher power density. Such elevated power densities approach the boundaries of conventional air-cooling capabilities, requiring more advanced thermal management solutions. In this study, we define the power density of the AI data center as follows:

$$P_{area} = P_{rack} \cdot N_{rack} \tag{S1}$$

P_{area}, P_{rack}, and N_{rack} represent the rated power density of the AI data center per area, the rated power density of the AI server racks, and the rack number per data center area, respectively. Specifically, P_{rack} is set as 50kW per rack, which is an identified maximum value for commonly used data center air-cooled systems [1, 2]. Notably, this threshold has already been reached by the Nvidia DGX B200 system, which employs a four-systems-per-rack design, each with a rated power of 14.2 kW. To align with industry-standard configurations that accommodate high-power AI racks, the rack density N_{rack} is further set as five racks per 9.29 m^2 (100 square feet) to align with the common industry design patterns for supporting increased high energy density AI racks [3, 4]. Based on these parameters, the resulting power density of the AI data center is approximately 26,910 W/m².

Table S1. *The name, time scope, served computing task, and hardware structure of collected AI traces*

Name	Time Scope	Computing Tasks	Server Structure	Reference
Earth	04/01/2020 – 09/28/2020	Majorly deep learning training	1144 Volta GPUs	Helios[5]
Saturn	04/01/2020 – 09/28/2020	Majorly deep learning training	2096 Pascal & Volta GPUs	Helios[5]
Uranus	04/01/2020 – 09/28/2020	Majorly deep learning training	2112 Pascal GPUs	Helios[5]
Venus	04/01/2020 – 09/28/2020	Majorly deep learning training	1064 Volta GPUs	Helios[5]
PAI2020	07/01/2020 – 08/31/2020	AI training and inference workloads	6742 Tesla, Pascal, & Volta GPUs	Alibaba PAI[6]
Kalos	03/01/2023 – 08/31/2023	Large language model training and inference	2416 Ampere GPUs	Acme[7]

Seren	03/01/2023 – 08/31/2023	Large language model training and inference	2288 Ampere GPUs	Acme[7]

The workloads for the defined AI data center are constructed using real-world operational data. Specifically, seven publicly available AI workload traces from existing AI data centers were collected, representing the most comprehensive and high-quality open-source datasets available to our knowledge. Table S1 summarizes key details of these traces, including the trace name, time scope, types of computing tasks served, hardware configurations, and data sources. To the best of our knowledge, these datasets represent all publicly available AI computing workload traces from real-world GPU clusters within the past five years. As illustrated in Table S1, these AI traces vary in the number and types of GPUs utilized, as well as the specific computing tasks performed. The resulting server utilization rates derived from these traces are presented in Figure S1. Additional detailed characteristics of each AI workload trace can be found in the references listed within the table. It should be noted that this analysis could be further expanded and refined as additional operational data becomes publicly available, particularly data from state-of-the-art AI clusters. To facilitate a comprehensive year-long simulation, the time scope of all AI traces is extended to one year by repeating the available data. These traces provide minute-level GPU usage metrics for processing all requested jobs. The collected data are then integrated into the power utilization rate using the following equation:

$$u_{power} = \frac{N_{GPU,active}}{N_{GPU,total}} \cdot \bar{u}_{active} \tag{S2}$$

u_{power}, $\bar{u}_{active}$, $N_{GPU,active}$, and $N_{GPU,total}$ represent the power utilization rate of all AI servers, the average power utilization rate of all active AI servers, the number of active GPUs, and the number of all GPUs, respectively. This formulation excludes CPU-based tasks in the traces, as AI servers are primarily driven by GPU workloads. Moreover, the $\bar{u}_{active}$ value is determined by the recently reported experimental data, set as 80% of the rated power of the AI server [8, 9]. Using this approach, all AI traces are converted into workload representations with power utilization rates ranging from 0 to 1. To facilitate efficient simulation and interaction with hourly grid dynamics, the minute-level data is aggregated into hourly averages. This not only enhances computational efficiency but also optimizes the application of the DRL-based approach. For a clearer visualization, Figure S1 illustrates the monthly average utilization across all AI workloads. Based on their utilization characteristics, workloads are categorized as follows: low-utilization (Kalos), medium-utilization (Earth, PAI2020, Seren), and high-utilization (Saturn, Uranus, Venus).

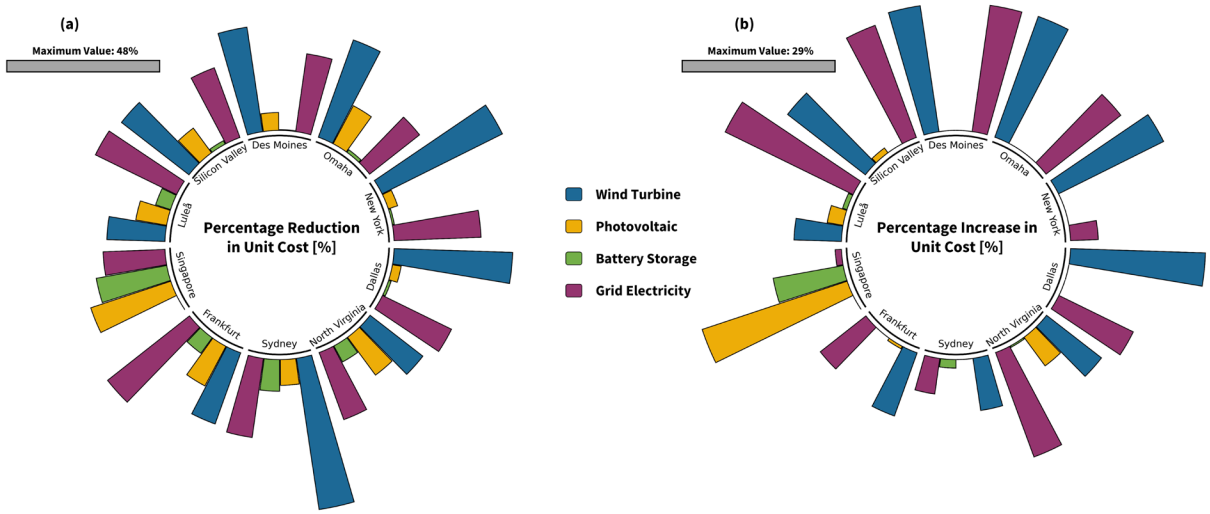

Figure S1. *The monthly average power utilization rate of all AI workloads.*

The cooling infrastructure and data center systems are developed based on a reference model established in prior research [10, 11]. To accurately simulate thermal behavior, the EnergyPlus model is employed, incorporating a server approach temperature table derived from computational fluid dynamics (CFD) simulations. Specifically, this data center model defines a rack space of 557 m^2, leading to a defined rated capacity of 15 MW for the AI data center by combining the energy intensity definitions. The cooling configuration consists of a waterside economizer with a cooling tower to leverage free cooling, supplemented by a chiller and a Computer Room Air Handler (CRAH) system to regulate the temperature of AI server racks efficiently. The design parameters for the cooling tower, chiller, pumps, and fans are adapted from previous studies [11], to ensure they meet the cooling requirements associated with the specified power density.

S2. Climate, Grid, and Renewable Energy Data Across the Ten Locations

This section provides data sources and values for supporting the cost-effective optimization approach in this study, as described in the **Integrated Data and Optimization Framework for Sustainability Solutions of AI Data Centers** Section. To ensure a comprehensive analysis, this study incorporates real-world climate, grid dynamics, and renewable energy cost data. The data sources for each location are listed in Table S2. Climate data, including temperature, humidity, solar irradiation, and wind speed, are sourced from the Global Typical Meteorological Years (TMYx) dataset, providing insight into climate variations across the selected locations. Grid dynamics data are collected from the most recent available year at the time of this study, with U.S. data from 2024 and data for other regions from 2023. Incorporating real-time

grid data allows the analysis to accurately capture regional variations in electricity pricing and grid carbon intensity, leading to a more precise evaluation of energy efficiency and sustainability. The climate and grid data are too large to be presented, which are available in our public repository https://github.com/PEESEgroup/AI-Data-Center-Analysis. Additionally, offshore wind turbines, solar panel, and energy storage system costs for each location are compiled from established sources listed in Table S2. The specific cost values of the offshore wind, solar, and battery storage cost values of each location are further presented in Table S3. Specifically, the yearly cost for battery storage is calculated based on a 15 year lifetime following the Annual Technology Baseline (ATB) scenario [12]. These estimates reflect utility-scale infrastructure, aligning with the large-scale energy demands of AI data centers. For on-site AI computing, only offshore wind turbines are considered. Energy generation for each location is simulated using EnergyPlus, where wind turbine and photovoltaic (PV) performance is modeled based on local climate data and ATB base year (2023) settings. This ensures a realistic representation of renewable energy integration within the study's framework.

Table S2. *Data sources used in this study, including climate data, grid carbon factor, grid electricity price, and cost data for wind, solar, and battery storage.*

Location	Climate Data	Grid Carbon Factor	Grid Price Data	Wind Cost Data	Solar Cost Data	Battery Storage Cost Data
Sydney	Global TMYx Data[13]	Electricity Map Tool[14], sourced from AEMO	AEMO data archive[15]	IEA LCOE Calculator[16]	IEA LCOE Calculator[16]	CSIRO annual GenCost report[17]
Frankfurt	Global TMYx Data[13]	Electricity Map Tool[14], sourced from ENTSO-E	SMARD wholesale market data[18]	IEA LCOE Calculator[16]	IEA LCOE Calculator[16]	IRENA Cost Report[19]
Singapore	Global TMYx Data[13]	Electricity Map Tool [14], sourced from EMA	EMC wholesale market data[20]	NREL Report[21]	NREL Report[21]	INOX Solar Market Analysis[22]
Luleå	Global TMYx Data[13]	Electricity Map Tool [14], sourced from ENTSO-E	Ember wholesale market data[23]	IEA LCOE Calculator[16]	IEA LCOE Calculator[16]	The Straits Times Report[24]
Silicon Valley	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]
Des Moines	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]
Omaha	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]
New York	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]
Dallas	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]
North Virginia	Global TMYx Data[13]	EIA 930 Real-time Monitor[25]	EIA Wholesale Market Data[26]	NERL SLOPE[27]	NERL SLOPE[27]	NREL ATB Baseline Data[12]

Table S3. *The offshore wind, solar, and battery storage yearly installation cost values of each location.*

Location	Wind Cost Value [$/MWh]	Solar Cost Value [$/MWh]	Battery Storage Cost [$/kWh]
Sydney	43	38	19
Frankfurt	53	63	57
Singapore	200	63	31
Luleå	44	40	25
Silicon Valley	47	41	32
Des Moines	32	45	32
Omaha	30	43	32
New York	37	53	32
Dallas	31	39	32
North Virginia	40	46	32

S3. Deep Reinforcement Learning for AI-focused Data Center Operation

This section provides additional information regarding the DRL approach introduced in the **Methodology Section** of the manuscript. Specifically, Figure S2 shows the workflow of the DRL simulation process and Algorithm S1 presents the policy training process of the DRL model.

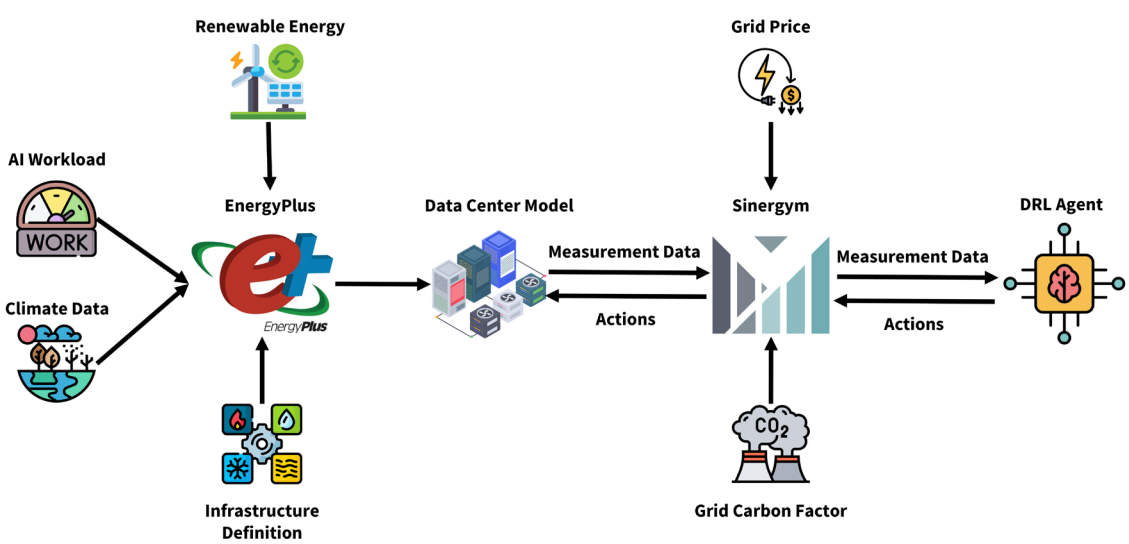

Figure S2. Workflow of data and software packages used in the DRL simulation process. This diagram illustrates the integration of input datasets and computational tools employed to develop, train, and evaluate the deep reinforcement learning model used in this study.

Algorithm S1. DRL policy training procedure

```
Input: AI data center environment, Total number of cases N
for each iteration (episode) do
    Choose a case c from N through randomly selecting a location
    state s ← Reset AI data center environment
    reward r ← 0
    for each environment step do
        Randomly select an AI server utilization rate among the defined range (25%, 75%)
        Choose action a based on the current state s using policy from the currently trained model
        Take action a on the AI data center environment and observe the next state s' and reward r
        R ← R + r
        Store (s, a, s', r) in Buffer
        If update do
            Update the neural network
        end if
    end for
end for
```

S4. Cost-efficient Optimization Problem Formulation

In this study, we applied an optimization-based approach to evaluate the maximum demand response potential and identify cost-optimal renewable energy installation strategies across all locations and AI workloads. Specifically, the maximum demand response potential is assessed by formulating the following optimization problem:

$$
\begin{aligned}
&\min_{\boldsymbol{P}_{ch},\boldsymbol{P}_{dis},\boldsymbol{P}_{grid},\boldsymbol{E}_{SoC}} \sum_{t=1}^{T} P_{grid,t}\cdot C_{grid,p,t} \\
&s.t. \quad P_{dem,t} = P_{grid,t} + P_{dis,t} - P_{ch,t}, \quad t\in T \\
&\qquad E_{SoC,t} = E_{SoC,t-1} - P_{dis,t} + P_{ch,t}, \quad t\in T \\
&\qquad E_{SoC,0} = E_{SoC,\min} \\
&\qquad E_{bat,\min} \le \mathrm{E}_{SoC,t} \le E_{bat,\max}, \quad t\in T \\
&\qquad 0 \le P_{ch,t} \le P_{bat,\max}, \quad t\in T \\
&\qquad 0 \le P_{dis,t} \le P_{bat,\max}, \quad t\in T
\end{aligned}
\tag{S3}
$$

where P_{ch}, P_{dis}, P_{grid}, and P_{dem} represent the battery charging power, battery discharging power, grid purchase power, and demand power of the AI data center obtained from the DRL controller, respectively; E_{SoC} is the state of charge (SoC) status of the battery, and the $E_{bat,min}$ and $E_{bat,max}$ are defined as 20% and 95% of the total capacity of the battery E_{bat}, respectively; $C_{grid,p}$ represents the grid price; t is the time index and ranges from 1 to 8760 for the whole year simulation; the maximum charging power and discharging power are defined as the total battery capacity divided by four hours ($P_{bat,max}$), reflecting the four-hour battery system used in this study. For carbon-aware demand response, the $C_{grid,p}$ should be replaced by the grid carbon factor $C_{grid,c}$.

Moreover, the cost-optimal renewable energy installation strategies are assessed by formulating the following optimization problem:

$$
\begin{aligned}
&\min_{\substack{\boldsymbol{P}_{ch},\boldsymbol{P}_{dis},\boldsymbol{P}_{grid},\boldsymbol{E}_{SoC},\\ E_{bat},P_{wind},P_{solar}}} \sum_{t=1}^{T} P_{grid,t}\cdot C_{grid,p,t} + E_{bat}\cdot C_{bat} + E_{wind}\cdot C_{wind} + E_{solar}\cdot C_{solar} \\
&s.t. \quad P_{dem,t} = P_{RE,t} + P_{grid,t} + P_{dis,t} - P_{ch,t}, \quad t\in T \\
&\qquad \sum_{t=1}^{T} P_{grid,t}\cdot C_{grid,c,t} \le R\cdot \sum_{t=1}^{T} \overline{P}_{dem,t}\cdot C_{grid,c,t}, \quad t\in T \\
&\qquad E_{wind}, E_{solar}, P_{RE,t} = F_{EP}\left(P_{wind}, P_{solar}\right), \quad t\in T \\
&\qquad E_{SoC,t} = E_{SoC,t-1} - P_{dis,t} + P_{ch,t}, \quad t\in T \\
&\qquad E_{SoC,0} = E_{bat,\min} \\
&\qquad E_{bat,\min} \le \mathrm{E}_{SoC,t} \le E_{bat,\max}, \quad t\in T \\
&\qquad 0 \le P_{ch,t} \le P_{bat,\max}, \quad t\in T \\
&\qquad 0 \le P_{dis,t} \le P_{bat,\max}, \quad t\in T \\
&\qquad E_{bat} \ge 0, P_{wind} \ge 0, P_{solar} \ge 0
\end{aligned}
\tag{S4}
$$

where $P_{RE,t}$, P_{wind}, and P_{solar} represent the renewable energy generation power at each time step, the installed wind power capacity, and the installed solar power capacity, respectively; E_{wind} and

E_{solar} are the generated total energy for the whole year simulation, which are calculated by the EnergyPlus model F_{EP}; C_{bat}, C_{wind}, and C_{solar} represent the cost values of wind, solar, and battery storage installations; $\bar{P}_{dem,t}$ is the demand (grid) power by using the baseline controller with no renewable energy and battery installations, and R represents the required reductions in total carbon emissions, which is set as 50% during the analysis. The data used in the above optimization problems has been introduced in previous texts.

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